

SWASTHIKA P S

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🌐 LinkedIn

🐙 GitHub

🌐 Portfolio

PROFILE

Computer Science undergraduate with hands-on experience in building scalable full-stack web applications. Skilled in developing responsive user interfaces, designing RESTful APIs, and deploying production-ready systems. Interested in solving real-world problems through efficient and user-centric web solutions.

EDUCATION

B.Tech in Computer Science and Engineering <i>Amrita Vishwa Vidyapeetham, Coimbatore</i>	Jun 2023 – Present CGPA: 8.37
Senior Secondary (CBSE) <i>Sri Chaitanya Techno School, Neelambur</i>	Jun 2021 – Apr 2023 XII: 90.6%
Secondary (CBSE) <i>Kathir Vidyaa Mandhir, Neelambur</i>	2020 – 2021 X: 93.8%

PROJECTS

LexFix – Accessibility-First Language Learning Platform

- Built a full-stack cloud-based language learning platform designed for neurodivergent learners with adaptive multi-modal learning (text, audio, visual).
- Developed an accessible and responsive UI focused on reducing cognitive load and improving user experience.
- Designed scalable backend architecture with REST APIs and role-based access control for learners, parents, and educators.
- Integrated real-time collaboration features and intelligent feedback for pronunciation and language learning.
- Implemented secure data handling and deployed the application on a cloud platform for production use.
- **Tech Stack:** Next.js, Node.js (Express), PostgreSQL, MongoDB, Hugging Face (wav2vec2), spaCy

Hall Booking Portal

- Developed a full-stack booking system with real-time conflict detection and scheduling validation.
- Implemented secure authentication and role-based dashboards for Admin and Users.
- Designed RESTful APIs for booking, authentication, and notification handling.
- Built responsive UI components and admin dashboard for booking analytics.
- **Tech Stack:** HTML, CSS, JavaScript, React, Node.js, Express, MongoDB

Secure Child Location Tracking System

- Developed secure backend system for real-time location tracking with encrypted data handling.
- Implemented multi-factor authentication using password and OTP (Twilio API).
- Applied AES-256 encryption and PBKDF2 hashing for secure data storage.
- Designed role-based access control for Parent, Child, and Admin users.
- Built scalable APIs for secure frontend-backend communication.
- **Tech Stack:** React, Node.js, Express.js, MongoDB

SKILLS

Programming Languages: C++, Python, C, SQL

Web Development: HTML, CSS, JavaScript, React

Databases: MongoDB

Mobile Development: Flutter

Machine Learning: Regression, SVM, Random Forest, Clustering

Core CS: Data Structures and Algorithms, OOP, DBMS, OS, Computer Networks

EXTRACURRICULAR ACTIVITIES & INTERESTS

- Interest in chess, with multiple tournament wins, including a district-level championship and a Sahodaya championship (2017)

- Interest in logical reasoning, Sudoku, and problem-solving puzzles
- Active participant on HackerRank for coding practice

LANGUAGES

Tamil (Native) — English (Proficient)